
ME 309: Theory of Machines

Course Code	ME 309
Course Name	Theory of Machines
L-T-P-C	3-1-0-4
Prerequisites	Mechanics of Solids
Intended for	B.Tech.
Distribution	Core
Approval	3rd Ad-hoc Senate (Nov 2012)

Course Contents

- Kinematic pair, diagrams and inversion. Mobility and range of movement [5 Lectures]
- Displacement velocity and acceleration analysis of planar linkages [10 Lectures]
- Dimensional synthesis for motion, path and function generation. [8 Lectures]
- Cam profile synthesis [5 Lectures]
- Gears [10 Lectures]
- Dynamic force analysis [4 Lectures]
- Flywheel [3 Lectures]
- Inertia forces and balancing for rotating and reciprocating machines [12 Lectures]

Text Books

1. A. Ghosh, A. K. Mallik, **Theory of Mechanisms and Machines**, East West Press Pvt. Ltd.
2. Uicker, J. J., Shigley, J. E., and Pennock, G. R., **Theory of Machines and Mechanisms**, Oxford University Press.
3. Thomas Bevan, **Theory of Machines**, 4th Edition, Pearson
4. C. E. Wilson, J. P. Sadler, **Kinematics and Dynamics of Machinery**, Pearson.
5. R. L. Norton, **Design of Machinery**, McGraw Hill Company.